WORKED SOLUTION

STEP III 1990, Question 12

Mechanics • Particle on Moving Wedge

STEP III 1990, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q12

The key idea

Two things are conserved while the particle slides: horizontal momentum (the table is smooth, so there are no external horizontal forces) and energy (every surface is smooth). These two laws, together with the fact that the particle always moves along the face AC *relative to the wedge*, fix every velocity. After that, the bounce and the flight are just projectile motion while the wedge drifts at constant speed.

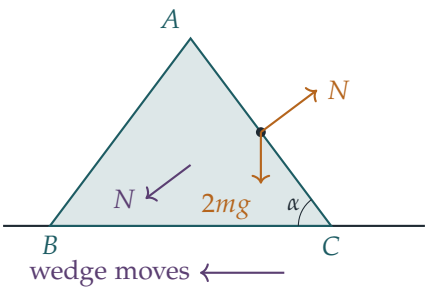
Solution

Setting up

Model the particle as a point mass $2m$ and treat the wedge as a rigid body of mass m. All contacts are smooth. Work in the vertical cross-section through A, B, C. Put B at the origin, with $C = (6a, 0)$ and $A = (3a, 4a)$. The height of A is $4a$ because the triangle is isosceles with $AB = AC = 5a$ and half-base $3a$. Let α be the angle ACB. Then

$$\sin\alpha = \tfrac{4}{5}, \qquad \cos\alpha = \tfrac{3}{5}.$$

The particle slides down AC, moving in the positive x-direction, and the wedge recoils in the negative x-direction.



Let u be the speed of the particle *relative to the wedge*, directed down AC. Let V be the

wedge's velocity, which is horizontal. The particle's velocity is the wedge's velocity plus its velocity relative to the wedge:

$$\mathbf{v}_P = \big(V + u\cos\alpha,\ -u\sin\alpha\big) = \big(V + \tfrac{3}{5}u,\ -\tfrac{4}{5}u\big).$$

Momentum and energy during the slide

The only external forces are gravity and the vertical reactions of the table, so horizontal momentum is conserved. It starts at zero, so

$$2m\big(V + \tfrac{3}{5}u\big) + mV = 0 \quad\Longrightarrow\quad V = -\tfrac{2}{5}u.$$

The particle's velocity is therefore $\big(\tfrac{1}{5}u,\ -\tfrac{4}{5}u\big)$ and the wedge's is $-\tfrac{2}{5}u$.

? **Ask yourself** Why can't we just conserve energy for the particle on its own?

The normal reaction on the particle is perpendicular to the face AC, but the face is moving. So the particle's actual velocity is not along AC, and N does work on it. Only the *system* conserves energy, because the pair of normal forces does zero total work.

The particle falls a height $4a$ by the time it reaches the bottom of AC. Energy for the whole system gives

$$\begin{aligned}
\tfrac{1}{2}(2m)\Big(\tfrac{1}{25} + \tfrac{16}{25}\Big)u^2 + \tfrac{1}{2}m\cdot\tfrac{4}{25}u^2 &= 2mg\cdot 4a \\
\tfrac{19}{25}u^2 &= 8ga \\
u^2 &= \frac{200ga}{19}.
\end{aligned}$$

The particle stays on the face throughout. Resolving with accelerations shows that the normal reaction is the constant $N = \tfrac{10}{19}mg > 0$.

Where everything is at the impact

Momentum is zero throughout, so the centre of mass of the system has no horizontal motion. Let the particle move x_P and the wedge x_W horizontally by the time the particle reaches the table. Then $2m\,x_P + m\,x_W = 0$. Relative to the wedge, the particle has gone from A to C, a horizontal distance of $3a$, so $x_P - x_W = 3a$. Hence

$$x_P = a, \qquad x_W = -2a.$$

At that instant the particle is at $x = 3a + a = 4a$ and the corner C is at $6a - 2a = 4a$. They are at the same place, as they must be, since the particle has just reached C.

The bounce and the flight

The table is fixed and smooth, and the impact is perfectly elastic. So the particle's horizontal velocity is unchanged and its vertical velocity is reversed. It leaves the table

with velocity $\left(\frac{1}{5}u,\ \frac{4}{5}u\right)$. Its time of flight back to the table is

$$T = \frac{2 \cdot \frac{4}{5}u}{g} = \frac{8u}{5g}.$$

During the flight it moves horizontally a distance $\frac{1}{5}u \cdot T = \dfrac{8u^2}{25g}$ to the right.

Once the particle has left it, the wedge has no horizontal force on it, so it carries on at speed $\frac{2}{5}u$ to the left. In time T it moves $\frac{2}{5}u \cdot T = \dfrac{16u^2}{25g}$. The particle is moving right and the wedge left, so the particle cannot meet the wedge again.

The distance between D and E is the sum of these two distances, because both started from $x = 4a$ and they moved in opposite directions:

$$DE = \frac{8u^2}{25g} + \frac{16u^2}{25g} = \frac{24u^2}{25g} = \frac{24}{25} \cdot \frac{200a}{19} = \frac{192}{19}a,$$

as required.

Checks

- Total kinetic energy at the impact: $\frac{19}{25}mu^2 = \frac{19}{25} \cdot \frac{200}{19}mga = 8mga$. This equals the potential energy lost.
- Momentum: $2m \cdot \frac{1}{5}u + m \cdot (-\frac{2}{5}u) = 0$.
- Stepping the motion numerically with $a = g = m = 1$ gives $DE \approx 10.1053$, and $\frac{192}{19} = 10.10526\ldots$.

Common slips

- Taking the particle's velocity to be along AC. That is only its velocity *relative to* the moving wedge.
- Writing $\frac{1}{2}(2m)u^2 = 2mg \cdot 4a$ for the particle on its own. This ignores the wedge's kinetic energy and the work done by N.
- Measuring DE from the original position of C rather than from the impact point $x = 4a$, where both the particle and C are at that moment.

Beyond STEP

A wedge sliding on a smooth table is a classic example of a system with *constraints*. Lagrangian mechanics handles such systems by choosing coordinates (here, the wedge's position and the particle's distance down the slope) so that the constraint forces never need to appear. The conserved horizontal momentum then comes from the symmetry

of the problem under horizontal translation, which is an example of Noether's theorem. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers moving frames and momentum conservation in multi-body systems.

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